

Stability and Control in Power Systems

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Abstract—The article focuses on different aspects that influence electric power systems, focusing on the performance and stability of synchronous generators, the role of system inertia, as well as the regulation of active and reactive power in modern grid conditions. With the utilization of the MATLAB Simulink software, the research aims to deepen understanding of the interactions that characterize electric energy systems.

An analysis was performed on how synchronous generators reacted to varying load types and different power factors, to the impact of system inertia on frequency stability, and to the influence of droop control in active power and frequency regulation. The analysis also examined the disturbances caused by reduced inertia. As more renewable energy sources are integrated into the grid, overall system inertia decreases, leading to more frequent and pronounced frequency variations. This highlights the need for additional resources to mitigate these fluctuations effectively.

Converter interfaced generation shows promise in enhancing grid stability, through coordinated active and reactive power control. The results offer insights into the design and operation of resilient, sustainable electric power systems.

I. INTRODUCTION

This research is divided into four sections, each addressing particular system behavior and performance of the generator at different conditions. The first part qualitatively captures the generator performance when the system is connected to a load with no other external stabilization available. The second part is aimed at a further understanding of the generator's behavior where a three-phase voltage source is used and where the performance of the generator is stabilized.

In the third part, crowed control and droop control contribution to the frequency stabilization is analyzed. To further increase the understanding of the usage of droop control, practical scenarios like changing the resistive load while a generator is actively working are examined to determine the level of disturbance droop control can equilibrate. Lastly, was studied the result of the Voltage Source Converter (VSC) on the balancing problem of the generator power output, which leads to disturbances in the system being simulated.

All the simulations employ MATLAB Simulink with the purpose to provide detailed and profound analyses of the responses of the system and control schemes employed.

II. TASK 1 AND 2

A simulation of a generator operating with a load has been developed. The simulation setup includes a synchronous generator which is connected to a load through a busbar, and the performance of the generator is controlled through the

mechanical power input into its turbine. The busbar enables a midpoint for measuring voltage, and the inductance to the active and reactive power flows. The load is designed by a combination of resistive, inductive, and capacitive elements with variable magnitude to meet different prescribed operating conditions.

In the case of this study, the generator was set at a mechanical power of 0.350 power units (p.u.), while the load was set to 500 MW which is 50% of the generator's nominal power.

When a purely resistive load was connected, it was noticeable that the generator only delivered active power, and there was no reactive power delivered. Also, it was apparent that during the simulation, there was no change in the system's voltage, showing that the system was stable. This tendency emphasizes the cyclic nature of their behavior as they are not reactive to the systems reactive power.

Power out of the generator under this condition is illustrated in Figure 1, where the yellow line shows active power (P) while reactive power (Q) is illustrated by the blue line. The data shows that the output of active power is equal to the load demand and reactive power is not relevant. This analysis establishes a steppingstone for further investigation of more complicated load arrangements, such as the presence of inductive and capacitive loads, and the subsequent influence on system stability.

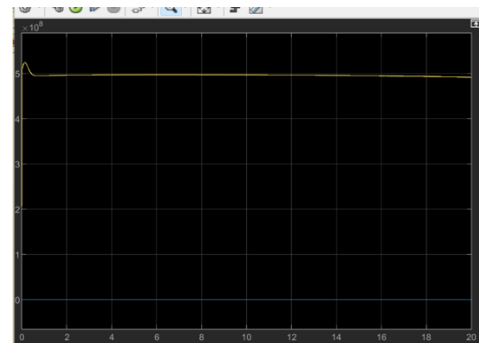


Fig. 1. Generator's Power

The electrical power provided can be seen in Figure 2.

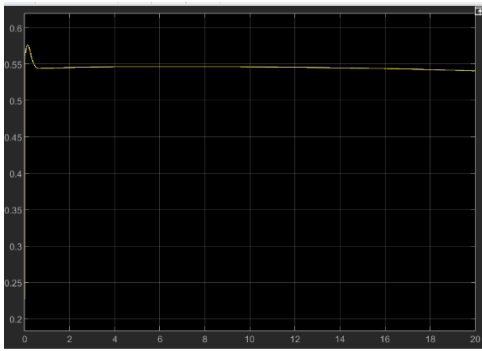


Fig. 2. Electrical power provided

In steady-state, it can be observed that the generator's electrical power is equal to the load demand, with slight variations (5×10^8 W vs. 0.55 p.u.) explained by the resistive component of the load. When connecting an inductive load (70% resistive, 30% inductive), the generator supplies both active and reactive power. This doesn't lead to a voltage drop in the generator's terminal voltage, as this sometimes happens when connecting an inductive load.

It is important to note that the higher reactive power demand may stress the generator, potentially leading to increased heating and potential inefficiency. [1]

The generator power can be seen in Figure 3.

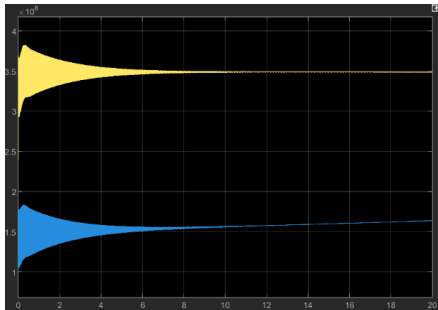


Fig. 3. Generator's Power

The electrical power provided can be seen in the following Figure 4.

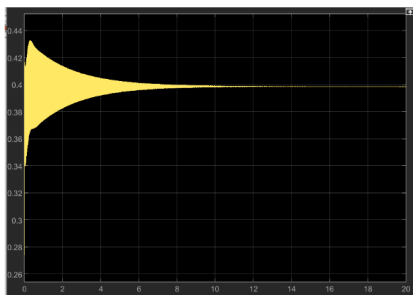


Fig. 4. Electrical power provided

When connecting a capacitive load (70% resistive and 30% capacitive), we observe that the generator absorbs reactive

power, additionally to providing active power. This is the opposite of what happens when the generator is connected to an inductive load, where it supplies reactive power. This doesn't lead to a voltage increase in the generator's terminal voltage, as this sometimes happens when connecting a capacitive load.

The generator power can be seen in Figure 5.

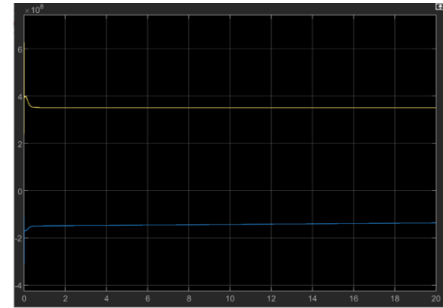


Fig. 5. Generator's power

Zoom on active power:

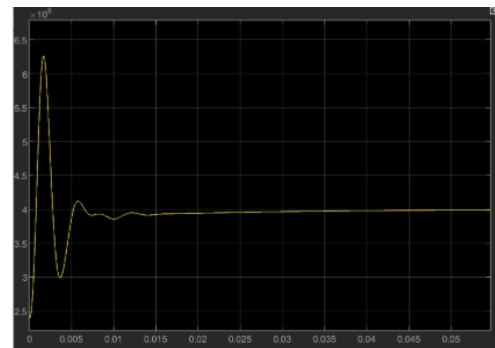


Fig. 6. Active Power

The electrical power provided can be seen in Figure 7.



Fig. 7. Electrical power provided

To analyze the role of a stronger power system in mitigating perturbations, it is first important to observe the behavior of the primary power system, with a perturbation load connected and without the influence of the other system.

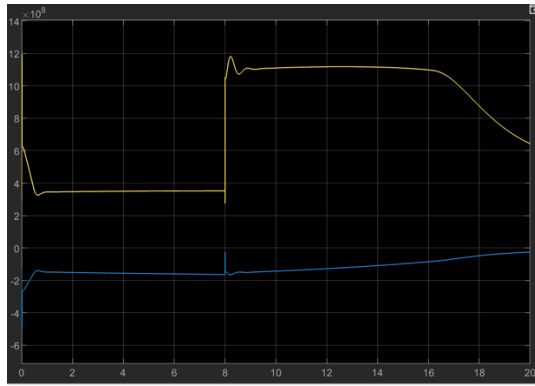


Fig. 8. Introduction of disturbance

When a disturbance is introduced, the primary system struggles to regulate the perturbation, requiring significant time to return to its pre-disturbance equilibrium. After introducing the three phases source, we can observe that the system is compensating way better (Figure 9), thanks to the three-phases circuits coming in hand to support the generator and enabling faster stabilization and improved overall system performance during disturbances.

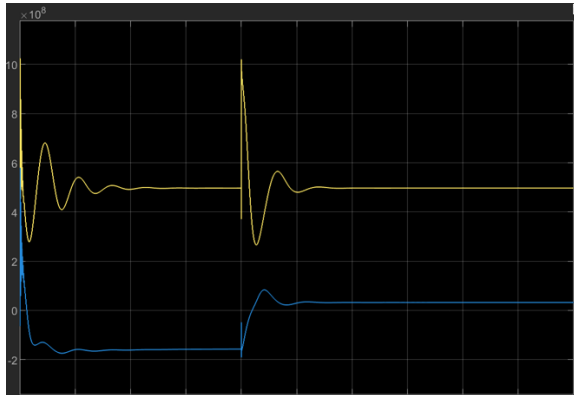


Fig. 9. System with three-phase voltage source

III. TASK 3

The system configuration includes a base load of 300 MW active power and 50 MVar reactive power, with an additional load of 40% of the generator's nominal power (400 MW) connected during the simulation. In this configuration, the generator's inertia was changed from 6s to 0.5s in small steps. It was observed that the frequency of the systems drops much quicker when the generator has a small inertia rather than when the generator has a bigger inertia, as seen in Figure 10.

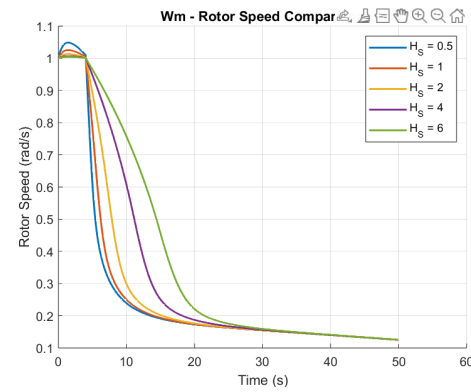


Fig. 10. Rotor speed comparison

The quicker frequency drops with lower inertia is due to the fact that the electrical power provided by the generator drops very quickly when the generator's inertia is low. In contrast, the electrical power provided by the generator with higher inertia maintains the electrical output for a much longer period of time, enabling the system to remain stable for extended periods. It was therefore observed that high inertia acts as a "buffer" against frequency disturbances, absorbing and mitigating their effects to preserve system stability. [2]

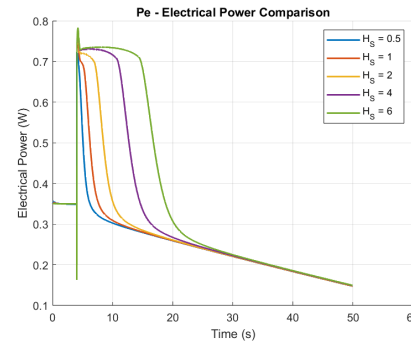


Fig. 11. Electrical power comparison

In this case, the mechanical power doesn't change.

Secondly, a simulation of a variable residential + commercial + industrial load as a random value around average power consumption was implemented in the model, keeping the same configuration as before.

Higher inertia serves as an effective "buffer" against frequency disturbances, reducing the magnitude of frequency changes. [2] When the generator has high inertia, the frequency remains more stable compared to scenarios with lower inertia, where frequency fluctuations are significantly more pronounced.

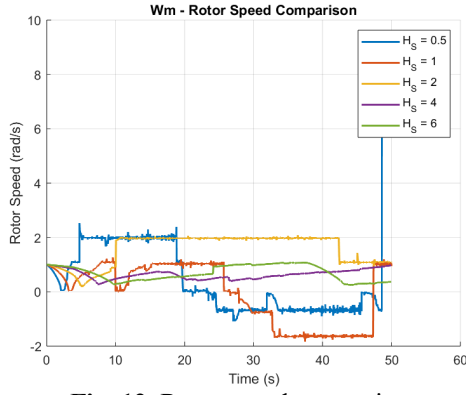


Fig. 12. Rotor speed comparison

It was also seen that the exciter plays an essential role in frequency regulation, as without it, the frequency simply goes up without being able to stabilize, highlighting its essential role in regulation. The exciter thus helps the system stay stable for as long as possible, and makes the system come back to a stable state whenever it is unstable, by regulating the electric field voltage effectively. [3] This can be seen in the following figures.

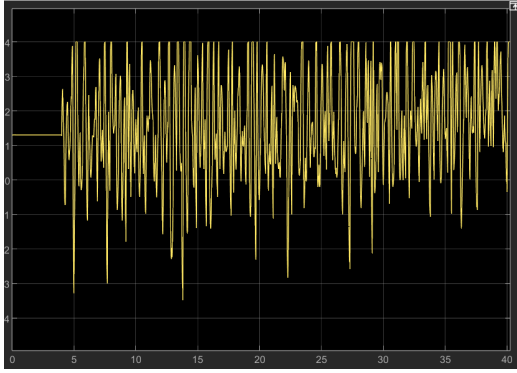


Fig. 13. Reactive power

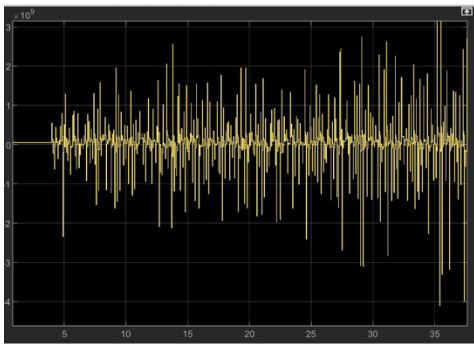


Fig. 14. Electric field voltage

IV. TASK 4

In this task, the main objective was to observe how the system behaves to the addition of a droop control system, and how inertia influences its performance. The initial analysis focused on the system operating without the droop control

mechanism, serving as a baseline to compare the stabilizing effects of droop control under varying inertia conditions.

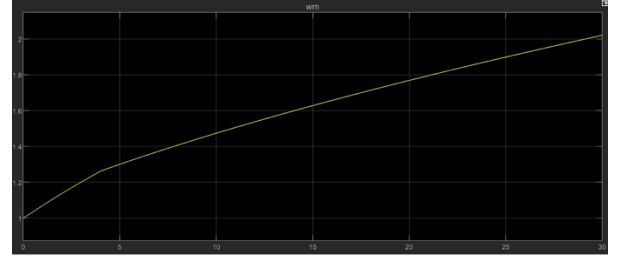


Fig. 15. System without droop correction

Without droop control, the generator is unable to stabilize the frequency, resulting in a significant frequency rise, reaching values nearly double the nominal operating frequency. Such instability would likely lead to catastrophic consequences, including generator failure and severe grid disruptions. [4]

The droop control is in fact able to stabilize the frequency by taking an average inertia value of 3. This configuration highlights the capability of droop control to maintain frequency stability and mitigate the risk of such extreme frequency deviations.

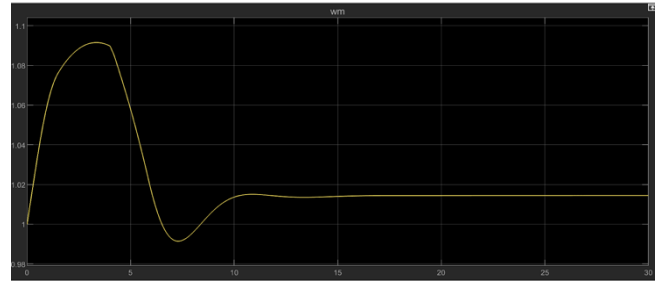


Fig. 16. System with droop correction

The droop control system effectively stabilizes the frequency, gradually restoring it to its nominal value. In a scenario where another generator is connected to the grid, the frequency regulation would be further improved due to shared load-balancing capabilities. [5] To better understand the role of inertia, the system was analyzed under varying conditions, starting with a higher inertia value of 6, to evaluate its influence on frequency stability.

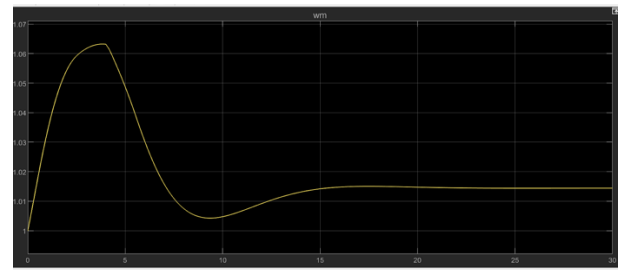


Fig. 17. System with droop correction and high generator inertia

With a higher inertia value of 6, the frequency changes become slightly smoother than it was at 3, and there are less fluctuations. This observation suggests that higher inertia enhances the effectiveness of the droop control system, leading to more stable frequency regulation. To further confirm this, the system was tested with a significantly lower inertia value of 0.5s to observe its impact on stability.

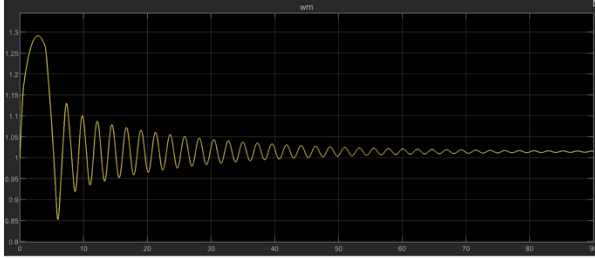


Fig. 18. System with droop correction and low generator inertia

As expected, with low inertia (0.5 s), the droop control system struggles to restore frequency stability effectively. This can be illustrated with the blades of a helicopter: let's take one case in which the blades are spinning at full speed. Throwing a tennis ball on it will slightly impact the rotary speed of the blades, and the system will be able to get back to full speed in no time. On the other hand, if a tennis ball was launched on the blades while they were at a low speed, the blades will take more time to recover from the hit to get the momentum back.

V. TASK 5

Simulations were then performed with a Voltage Source Converter (VSC) connected to the synchronous generator and a resistive load of 50% of the generator's nominal power. Midway through the simulation, an additional inductive load (20% of the generator's nominal power, 0.9 power factor) was connected.

Following the load connection, the generator's active and reactive power rose, fluctuating before stabilizing at a steady-state value. Similarly, the active and reactive power of the VSC also fluctuated after the load connection. The VSC, however, reduced the load on the generator by supplying part of the required amount of power. While the VSC's active power contribution decreased over time as the system stabilizes, it provided more power after the additional load was connected, thereby enhancing the generator's stability and overall system performance.

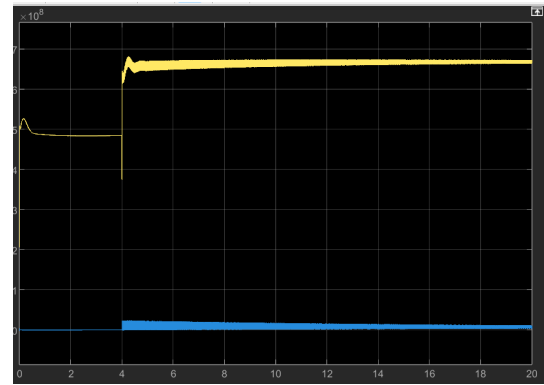


Fig. 19. Active and reactive power of synchronous generator

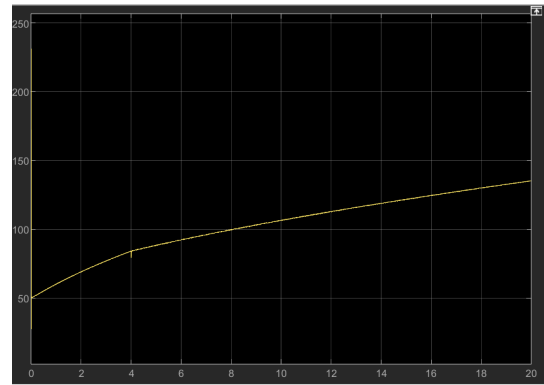


Fig. 20. Frequency of the Voltage Source Converter

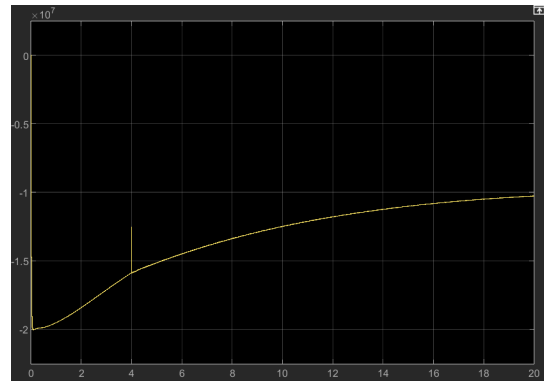


Fig. 21. Active power provided by the VSC

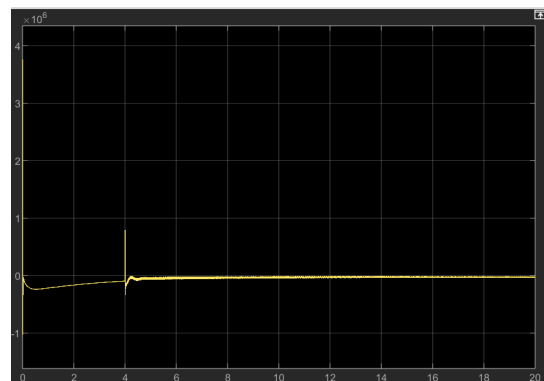


Fig. 22. Reactive power provided by the VSC

However, the system remains unstable, even if the VSC helps the generator provide power. In fact, the frequency is not regulated at all and keeps going in an upwards trend.

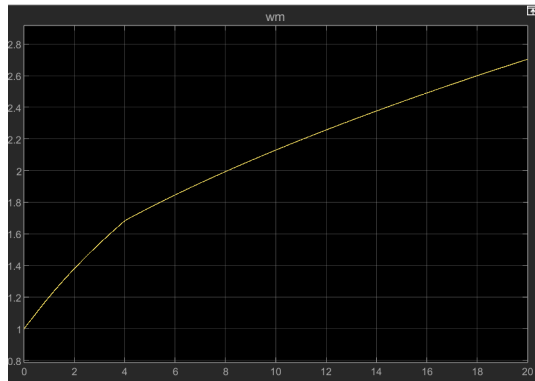


Fig. 23. Frequency of system

VI. TASK 6

The introduction of droop control significantly improved the system's ability to self-regulate the frequency, addressing the instability observed in Task 5. With an inertia value of 6 (as used previously), the droop control effectively stabilized the system. The frequency of the Voltage Source Converter (VSC), which was previously escalating uncontrollably, is now maintained within a stable range, demonstrating the effectiveness of droop control in mitigating frequency disturbances and enhancing overall system stability.

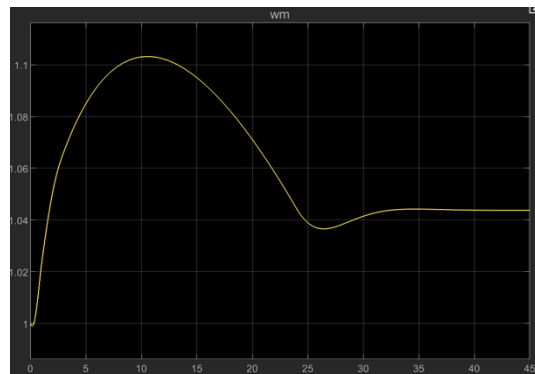


Fig. 24. System frequency with 6s inertia

The droop control mechanism effectively stabilizes the VSC frequency, bringing it smoothly closer to the nominal value of 50 Hz.

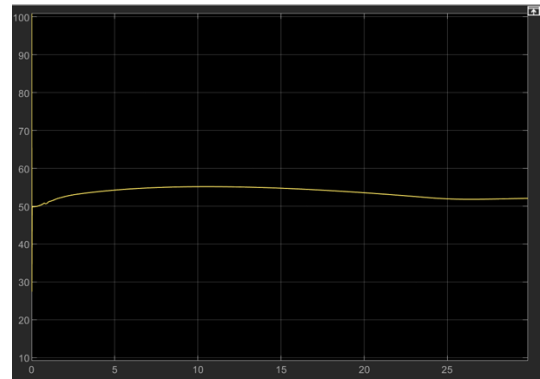


Fig. 25. Frequency of the Voltage Source Converter with 6s inertia

Additionally, the active power of the VSC also stabilizes and doesn't drop down after the droop regulation taking over.

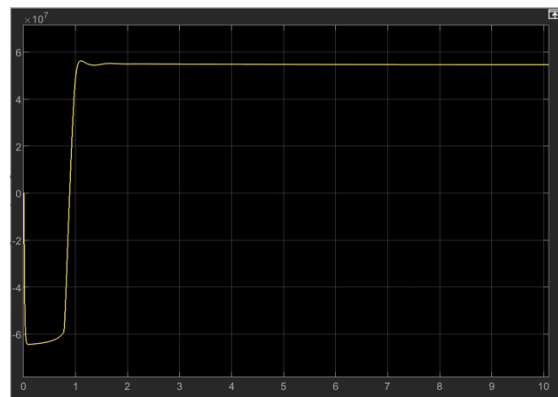


Fig. 26. Active power provided by the VSC

As shown, the value continuously corrected to remain stable. By dropping the inertia to 0.5 seconds, as explained in task 3, the system struggles to balance the frequency correctly.

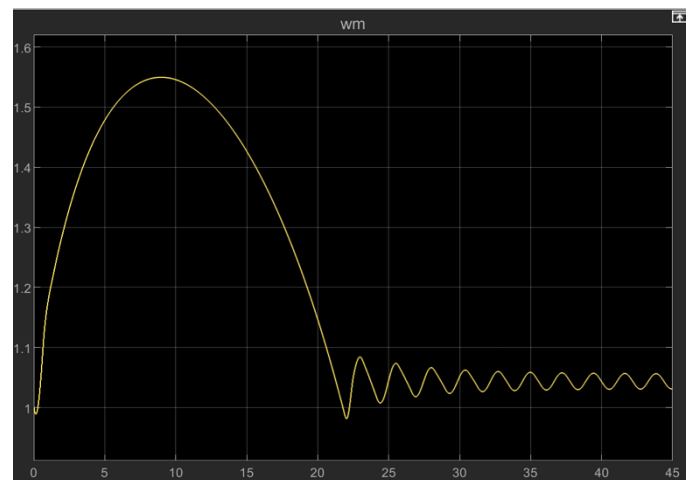


Fig. 27. System frequency with 0.5s inertia

It is thus confirmed that the more inertia the system has, the more stable it becomes.

An observation from Task 6 is that the droop control mechanism plays a crucial role in stabilizing the frequencies of both the generator and the Voltage Source Converter (VSC). By

coordinating the frequency regulation between these components, droop control ensures system-wide stability, effectively mitigating fluctuations and maintaining operational balance. [4]

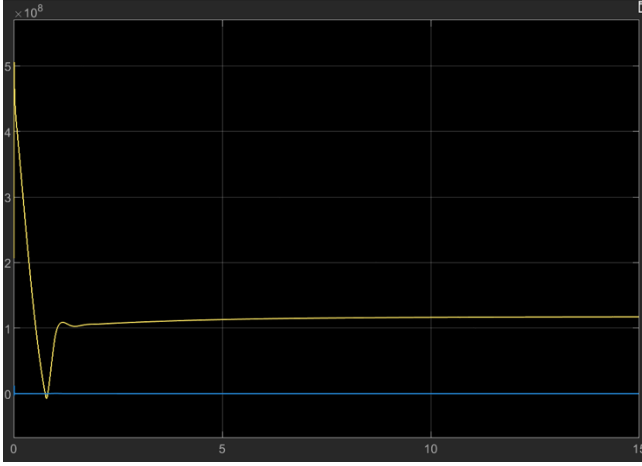


Fig. 28. Active and reactive power provided by generator

VII. CONCLUSION

This study explores the fundamental theory of machine parameters and control techniques in enhancing the stability of grid systems. Highlighted here are inertia, which is the main parameter in lowering frequency fluctuations; the efficacy of the droop control technique in stabilizing both the generator and the Voltage Source Converter (VSC); and, lastly, how VSCs enable demand-side management, which reduces the power demand on generators. These observations, in turn, raise the question of how to guarantee that modern grid operation remains reliable through adaptive strategies.

In light of current debates on integrating renewables into the energy mix and their implications for grid stability, this study underscores the urgency of addressing challenges related to declining system inertia and variable power generation. Modern grid measures, such as the deployment of advanced grid-scale batteries, flexible demand-side solutions, and enhanced digital monitoring tools, are critical in this context. Prospective analysis could deal with innovative control strategies, such as automatic droop control or predictive algorithms, which improve stability under disordered load and generation levels. Furthermore, investigating the impact of natural energy sources on inertia and power system dynamics would contribute to designing more sustainable, resilient, and efficient networks, ensuring that renewable integration supports, rather than destabilizes, the evolving energy grid.

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